

Amplitude Amplification: A Comprehensive Survey from Geometric Foundations to Algebraic Unification

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This survey traces the evolution of quantum amplitude amplification from its geometric origins in Grover’s algorithm to its algebraic formulation within the Quantum Singular Value Transformation (QSVT) framework. We develop a unified mathematical formalism intended to reduce the historical fragmentation of these algorithms. By establishing a consistent notational foundation early on, framing unstructured search as rotation within an invariant two-dimensional subspace, and following the progression through oblivious, fixed-point, and distributed amplification schemes, we aim to clarify how these methods can be interpreted within the broader language of singular-value polynomial synthesis.

I. INTRODUCTION

The discovery of quantum algorithms capable of outperforming their classical counterparts stands as one of the hallmark achievements of modern theoretical computer science. Among the fundamental primitives that enable such speedups, quantum amplitude amplification represents arguably the most versatile and ubiquitous. First introduced in its discrete, unweighted form by Lov Grover as a solution to the unstructured database search problem [1], the core mechanism has since undergone successive and profound generalizations. What began as a specific spatial rotation for finding marked items has matured into an encompassing algebraic framework capable of arbitrary structural manipulations over an N -dimensional Hilbert space [6].

Despite its foundational importance, the pedagogical and review literature on amplitude amplification often documents the field’s progression as a collection of isolated algorithmic techniques, each accompanied by idiosyncratic notation and distinct mathematical models. This survey aims to mitigate that fragmentation by providing a unified, rigorous, and conceptually cohesive exposition. We trace the trajectory of amplitude amplification from basic geometric reflections to Quantum Singular Value Transformation (QSVT), with an emphasis on maintaining notational consistency across paradigms.

A. The Unstructured Search Paradigm

To ground the subsequent analysis, we fundamentally abstract the algorithmic objective as the isolation of a desired quantum state within a large, potentially intractable state space. Let $\mathcal{H}$ be an N -dimensional Hilbert space, serving as the computational arena over n qubits, such

that $N = 2^n$. We rigorously partition $\mathcal{H}$ into two mutually exhaustive and strictly orthogonal subspaces:

1. The **target subspace** (or “good” subspace) $\mathcal{H}_{\text{Good}}$, encompassing the span of all valid solution states.
2. The **orthogonal subspace** (or “bad” subspace) $\mathcal{H}_{\text{Bad}}$, encompassing all non-target components.

Accordingly, we define the corresponding orthogonal projection operators Π_{Good} and Π_{Bad} , satisfying $\Pi_{\text{Good}}^2 = \Pi_{\text{Good}}$ and $\Pi_{\text{Good}} + \Pi_{\text{Bad}} = I$, where I is the identity operator on $\mathcal{H}$.

The problem is initialized by the action of an efficiently preparable unitary algorithm $\mathcal{A}$, which acts on a fiducial zero state $|0\rangle^{\otimes n}$ to produce an initial coherent trial state $|\text{All}\rangle$:

$$|\text{All}\rangle = \mathcal{A}|0\rangle^{\otimes n}. \quad (1)$$

In the absence of any amplification technique, directly measuring $|\text{All}\rangle$ in the computational basis yields a target solution with an initial success probability p , defined strictly as the expectation value of the target projector:

$$p = \langle \text{All} | \Pi_{\text{Good}} | \text{All} \rangle = \|\Pi_{\text{Good}} |\text{All}\rangle\|^2. \quad (2)$$

When p is overwhelmingly small (e.g., $p \sim \mathcal{O}(1/N)$ in typical unstructured search scenarios), classical randomized sampling necessitates $\mathcal{O}(1/p)$ queries to succeed with high probability. Quantum amplitude amplification seeks to coherently boost this success probability to order unity using only $\mathcal{O}(1/\sqrt{p})$ applications of the unitary $\mathcal{A}$ and its associated target-recognizing oracle, thereby yielding the familiar quadratic improvement in the oracle model.

B. The Invariant Subspace Formalism

To formalize the action of amplitude amplification computationally, it is useful to adopt a standard mathematical formalism that remains globally consistent, independent of whether the initial state is known, oblivious, or dispersed across variable runtimes.

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We decompose the initial trial state $|\text{All}\rangle$ explicitly along the orthogonal direct sum $\mathcal{H} = \mathcal{H}_{\text{Good}} \oplus \mathcal{H}_{\text{Bad}}$. We define the strictly normalized **target state** $|\text{Good}\rangle$ and **non-target state** $|\text{Bad}\rangle$ as:

$$|\text{Good}\rangle = \frac{1}{\sqrt{p}} \Pi_{\text{Good}} |\text{All}\rangle, \quad (3)$$

$$|\text{Bad}\rangle = \frac{1}{\sqrt{1-p}} \Pi_{\text{Bad}} |\text{All}\rangle. \quad (4)$$

By construction, these components satisfy absolute orthogonality, $\langle \text{Good} | \text{Bad} \rangle = 0$. Adopting a trigonometric parameterization dictated by the initial success probability scalar, $\sin^2(\theta_0) = p$, the initial state is mapped precisely onto a purely abstract two-dimensional basis:

$$|\text{All}\rangle = \sin(\theta_0) |\text{Good}\rangle + \cos(\theta_0) |\text{Bad}\rangle. \quad (5)$$

The plane spanned by $\{|\text{Bad}\rangle, |\text{Good}\rangle\}$ serves as the invariant geometric setting for amplitude amplification. One of the recurring themes of this survey is that the evolution of amplitude amplification, from Grover's discrete reflections to the generalized functional viewpoint underlying QSVT, can often be understood as a sequence of increasingly flexible polynomial transformations acting on invariant two-dimensional subspaces analogous to the one defined in Equation (5).

Symbol	Meaning / Standing Assumption
$\mathcal{H}_{\text{Good}}, \mathcal{H}_{\text{Bad}}$	target/non-target subspaces
$\Pi_{\text{Good}}, \Pi_{\text{Bad}}$	orthogonal projectors onto those subspaces
$ \text{All}\rangle = \mathcal{A} 0\rangle^{\otimes n}$	prepared input state before amplification
$p = \ \Pi_{\text{Good}} \text{All}\rangle\ ^2$	initial success probability
$N = 2^n, M$	search-space size and marked count
$\sin^2(\theta_0) = p$	angle parameter used across sections
Oracle model	complexity measured in oracle/query calls

TABLE I. Core notation and assumptions used throughout the survey.

C. Survey Roadmap

The remainder of this paper is structured to construct the conceptual edifice of amplitude amplification systematically:

Section II formalizes the unstructured search problem in the black-box setting, defines the oracle/query-cost model used throughout this survey, and states the asymptotic target and lower-bound landscape.

Section III establishes the historical timeline and scope of this survey, positioning each algorithmic development in the broader transition from geometric search to algebraic signal processing.

Section IV presents the canonical Grover algorithm, re-derived firmly within the two-subspace notation introduced above, highlighting the core geometrical reflection mechanism and the instability inherent in unitarity.

Section V generalizes this geometry into Fixed-Point Amplitude Amplification (FPAA), resolving the critical instability issue when the initial success probability p is strictly unknown, by relying on Chebyshev polynomial phase synthesis.

Section VI introduces Oblivious Amplitude Amplification, a primitive indispensable for embedding subroutines when reflections about a known initial state cannot be synthesized.

Section VII then studies Controlled Quantum Amplification, which converts detection-style search procedures into finding procedures without asymptotic overhead, especially in quantum-walk settings.

Sections VIII and IX address advanced modifications geared towards scaling factors, namely Fixed-Point Oblivious amplification and Distributed Amplitude Amplification, the latter formulated explicitly to mitigate structural bottlenecks in Noisy Intermediate-Scale Quantum (NISQ) devices.

Section X details Variable-Time Amplitude Amplification (VTAA), optimizing overhead when discrete sub-components terminate unpredictably.

Finally, **Section XI** places the preceding modules in the language of Quantum Singular Value Transformation. By subsuming the initial state preparation unitary into a broader block-encoding architecture, we show how the geometric intuition over $\{|\text{Bad}\rangle, |\text{Good}\rangle\}$ extends to an algebraic calculus over the singular values of a general matrix. **Sections XII, XIII, and XIV** then survey applications of this framework, discuss the polynomial lower bounds relevant to its optimality, and outline several directions for fault-tolerant and automated quantum algorithm design.

II. PROBLEM FORMULATION AND THE QUANTUM QUERY MODEL

We consider unstructured black-box search over the Boolean hypercube. Let

$$f : \{0, 1\}^n \rightarrow \{0, 1\}$$

be the indicator of marked inputs, with total search-space size $N = 2^n$. Define $\text{All} = \{0, 1\}^n$ and let

$$M = |f^{-1}(1)|$$

denote the number of marked items. A uniformly random classical guess succeeds with probability

$$p = \frac{M}{N}.$$

Oracle access is given by the standard bit-flip unitary $\mathcal{O}_f$ acting on an input register and one ancilla qubit:

$$\mathcal{O}_f |x\rangle |q\rangle = |x\rangle |q \oplus f(x)\rangle. \quad (6)$$

Preparing the ancilla in $|- \rangle = (|0 \rangle - |1 \rangle)/\sqrt{2}$ yields the equivalent phase-kickback form

$$\mathcal{O}_f |x \rangle |- \rangle = (-1)^{f(x)} |x \rangle |- \rangle. \quad (7)$$

For the remainder of this survey, we assume access to this induced phase oracle and write it as $O : |x \rangle \mapsto (-1)^{f(x)} |x \rangle$.

Complexity is measured in *query complexity*: the dominant cost is the number of oracle calls, while intermediate non-oracle unitary operations are treated as lower-order overhead unless explicitly stated otherwise.

Under this model, classical randomized search requires $\mathcal{O}(N/M)$ oracle queries to find a marked item with constant success probability, while amplitude amplification achieves $\mathcal{O}(\sqrt{N/M})$ [1, 9]. Crucially, Bennett, Bernstein, Brassard, and Vazirani proved that $\Omega(\sqrt{N/M})$ queries are necessary in the unstructured oracle model [3], so Grover-type scaling is asymptotically optimal. We return to the formal lower-bound mechanisms in Section XIII.

III. HISTORICAL BACKGROUND AND SURVEY SCOPE

The theoretical evolution of amplitude amplification is characterized by a steady abstraction from rigid geometric rotations into generalized algebraic transformations. Over the past three decades, algorithmic generalizations have systematically removed the operational constraints of the original unstructured search problem—such as the requirements of a known initial state, a known target density, or a monolithic coherent hardware architecture.

Table II outlines the chronological milestones that define this lineage.

Survey Scope: Rather than presenting these developments as isolated algorithmic tricks, a central objective of this survey is to reconstruct the mathematical bridge connecting them. We restrict our scope to the algorithmic lineage that carries standard geometric amplification into modern polynomial signal processing. By formalizing a single invariant subspace notation early on, we traverse fixed-point stabilization, oblivious generalizations, temporal variations, and spatial partitioning, and then relate these paradigms to Quantum Singular Value Transformation (QSVT).

IV. STANDARD AMPLITUDE AMPLIFICATION AND UNSTRUCTURED SEARCH

We now formalize the core geometric mechanism that underpins standard amplitude amplification, originally discovered by Lov Grover in 1996 for the unstructured database search problem [1]. The objective is to construct an iterative operator capable of monotonically rotating the initial state vector $|\text{All}\rangle$ toward the target subspace $\mathcal{H}_{\text{Good}}$, thereby driving the success probability toward

unity. While Grover’s original formulation relied strictly on the uniform superposition generated by the Walsh-Hadamard transform, it was subsequently rigorously generalized by Brassard, Høyer, Mosca, and Tapp [9] to accommodate any arbitrary state-preparation unitary $\mathcal{A}$, formally elevating the technique from a specific search algorithm to a universal amplitude amplification subroutine. In their operator formalism, the generalized iterate is written

$$Q = -\mathcal{A}S_0\mathcal{A}^{-1}S_\chi, \quad (8)$$

with $S_0 = I - 2|0\rangle^{\otimes n}\langle 0|^{\otimes n}$ and $S_\chi = I - 2\Pi_{\text{Good}}$ [9].

To maintain algebraic continuity with the unified formalism established in Section I, we derive the algorithm entirely within the invariant two-dimensional subspace spanned by $\{|\text{Bad}\rangle, |\text{Good}\rangle\}$, deliberately aligning the original unstructured problem mechanics with our generalized metric spaces.

A. Initialization and the Uniform Superposition

Because the database is completely unstructured, the algorithm initializes from the zero state $|0\rangle^{\otimes n}$ and applies the tensor-product Hadamard operation $H^{\otimes n}$ to explicitly define the initial trial state $|\text{All}\rangle$:

$$|\text{All}\rangle = H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{x \in \text{All}} |x\rangle. \quad (9)$$

We rigorously define the normalized geometric components representing the non-solutions and solutions across the entire database. If there are exactly M target solutions $\{|x \in \text{All} : f(x) = 1\}| = M$, we define the orthogonal bases:

$$|\text{Good}\rangle = \frac{1}{\sqrt{M}} \sum_{f(x)=1} |x\rangle, \quad (10)$$

$$|\text{Bad}\rangle = \frac{1}{\sqrt{N-M}} \sum_{f(x)=0} |x\rangle. \quad (11)$$

By direct superposition, the uniform initialization state partitions strictly into these two parameters:

$$|\text{All}\rangle = \cos(\theta/2) |\text{Bad}\rangle + \sin(\theta/2) |\text{Good}\rangle, \quad (12)$$

where $\sin(\theta/2) = \sqrt{M/N}$. Note that establishing the angle parameter as $\theta/2$ (as opposed to θ_0 in Section IB) intentionally yields a rotational step size of θ , a standard convention derived directly from Grover’s specific construction. The two parameterizations are equivalent under the relation $\theta = 2\theta_0$, with $p = \sin^2(\theta_0) = M/N$.

B. The Reflection Geometry: Oracle and Diffusion

With explicit respect to the initialization angle definitions, the action of the phase oracle O acts as a discrete

Year	Algorithmic Milestone	Primary Contribution
1996	Grover's Algorithm [1]	Unstructured search via geometric reflection
2000	Amplitude Estimation [9]	Extracting target probabilities via phase estimation
2002	General Amplitude Amp. [9]	Generalizing search to arbitrary unitary algorithms
2010	Variable-Time AA [11]	ℓ_2 -averaged scaling for early-halting branches
2014	Fixed-Point AA [15]	Monotonic convergence via Chebyshev polynomials
2017	Controlled Quantum Amplification [13]	Detection-to-finding conversion for walk-based search
2017	QAOA for Grover [14]	Near-optimal search using a single-qubit transverse-field mixer
2017	Qubitization & QSP [7, 8]	Optimal polynomial transforms on $SU(2)$ subspaces
2018	QSVT [6]	Algebraic unification via singular value transforms
2022	Fixed-Point Oblivious [16]	Resolving subnormalization via LCU damping
2025	Distributed AA [17]	Hardware-aware parallelization for NISQ devices
2026	Distributed Exact AA [18]	Exact distributed amplification for arbitrary amplitude distributions

TABLE II. Chronological development of the amplitude amplification framework, tracing the transition from basic geometric search primitives to generalized operator polynomials and modern distributed architectures.

Algorithm	Problem Solved	Representative Complexity
Grover / Standard AA	Unstructured search with known calibration	$\mathcal{O}\left(\sqrt{N/M}\right) = \mathcal{O}(1/\sqrt{p})$
FPA	Unknown lower-bounded p , fixed-point no-overshoot amplification	$\mathcal{O}\left(\frac{1}{\sqrt{p}} \log \frac{2}{\delta}\right)$
OAA	Subroutine amplification without direct input-state reflection	$\mathcal{O}(1/\sqrt{p})$
CQAA	Detection-to-finding conversion for abstract search operators	$\Theta(\text{QHT}(A, \text{init}\rangle))$
Fixed-Point OAA	Oblivious setting with robustness to unknown/drifted p	$\mathcal{O}(1/\sqrt{p})$ (up to constants)
VTAA	Variable stopping times across computational branches	$\mathcal{O}\left(\frac{1}{\sqrt{p}} \sqrt{\sum_j p_j T_j^2} \text{polylog}(T_m)\right)$
DEQAAA	Exact distributed amplification for arbitrary amplitude distributions	exact two-phase distributed depth

TABLE III. Survey-level comparison of core amplitude amplification variants.

reflection specifically about the $|\text{Bad}\rangle$ vector axis:

$$O|\text{All}\rangle = \cos(\theta/2)|\text{Bad}\rangle - \sin(\theta/2)|\text{Good}\rangle. \quad (13)$$

As an operator in projection notation, the oracle is most precisely written $O = I - 2\Pi_{\text{Good}}$, where Π_{Good} is the projector onto the target subspace $\mathcal{H}_{\text{Good}}$ established in Section I. This form is used throughout all subsequent sections.

The oracle O is systematically combined with another reflection operator R , constructed specifically to reflect about the uniform $|\text{All}\rangle$ initialization state:

$$R = 2|\text{All}\rangle\langle\text{All}| - I. \quad (14)$$

This structural reflection is implemented utilizing $H^{\otimes n}(2|0\rangle^{\otimes n}\langle 0|^{\otimes n} - I)H^{\otimes n}$. The intermediate operator $2|0\rangle^{\otimes n}\langle 0|^{\otimes n} - I$ acts as the complement of a multi-controlled Z operation, mechanically negating the sign when at least one computational bit evaluates to 1.

Together these operators define the complete Grover operator G :

$$G = RO. \quad (15)$$

With elementary rotation arithmetic, evaluating this composite operator precisely in the $\{|\text{Bad}\rangle, |\text{Good}\rangle\}$ basis yields the unitary matrix:

$$G = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (16)$$

That is, a vector deterministically rotates towards $|\text{Good}\rangle$ by a constant angle θ upon each sequential application of G . Evaluating the first application demonstrates this rotation explicitly:

$$G|\text{All}\rangle = \cos(3\theta/2)|\text{Bad}\rangle + \sin(3\theta/2)|\text{Good}\rangle. \quad (17)$$

After k repeated applications of the G operator, the state systematically evolves to:

$$\begin{aligned} |\psi_k\rangle &= G^k |\text{All}\rangle \\ &= \cos\left(\frac{(2k+1)\theta}{2}\right)|\text{Bad}\rangle \\ &\quad + \sin\left(\frac{(2k+1)\theta}{2}\right)|\text{Good}\rangle. \end{aligned} \quad (18)$$

The computational state $|\psi_k\rangle$ points entirely along the $|\text{Good}\rangle$ target axis when iteration numbers k and angular parameters θ satisfy the specific continuous alignment condition:

$$\frac{(2k+1)\theta}{2} = \frac{\pi}{2} \quad (19)$$

$$\implies k = \frac{\pi}{2\theta} - \frac{1}{2}. \quad (20)$$

Because k must be an integer, one chooses the nearest integer to this continuous optimum. In the standard

amplitude-amplification notation with $a = M/N$, a common choice is

$$k = \left\lfloor \frac{\pi}{4 \arcsin(\sqrt{M/N})} - \frac{1}{2} \right\rfloor, \quad (21)$$

which yields constant success probability [9]. In the sparse-marked regime $M \ll N$, one may use $\theta \approx 2\sqrt{M/N}$, giving the familiar asymptotic estimate

$$k \approx \frac{\pi}{4} \sqrt{\frac{N}{M}},$$

and hence the $\mathcal{O}(\sqrt{N/M})$ query scaling.

C. The High-Density Edge Case ($M > N/2$)

The exact geometric derivation of the optimum iteration count does not require a sparsity assumption; the condition in Equation (5) and the alignment equation above remain valid for all $0 < p < 1$. Sparsity ($M \ll N$) is used only to obtain the asymptotic estimate $k \approx (\pi/4)\sqrt{N/M}$ via the small-angle approximation. However, in scenarios where the exact solution density is exceptionally high ($M > N/2$), the initial separation angle $\theta/2$ evaluates broadly. Under these geometric conditions, a single computational application of the Grover iterate G can forcibly overshoot the target $|\text{Good}\rangle$ subspace, plunging the immediate success probability precipitously.

While classical random sampling already handles such high-density boundary problems efficiently, the coherent quantum setting requires some mitigation of this over-rotation. A standard device is to append a single auxiliary qubit, thereby doubling the effective database size $N \rightarrow 2N$ without introducing additional marked items. This dilution halves the relative target density to $M/2N \leq 1/2$, reducing the risk of immediate overshoot.

This vulnerability to overshooting is a consequence of the rigid unitary rotation structure. Avoiding such over-rotations when the target density is variable or unknown is one of the central motivations for Fixed-Point Amplitude Amplification.

D. QAOA-Inspired Search and the Transverse Field

An important 2017 development re-examined Grover search from the perspective of hardware-efficient alternating-operator circuits. Jiang, Rieffel, and Wang observed that Grover's diffusion operator is not only conceptually geometric but also hardware-intensive: implementing the reflection about $|\text{All}\rangle$ requires a multi-controlled operation whose standard circuit decomposition uses $\Theta(n)$ two-qubit gates [14]. This cost becomes a practical bottleneck on near-term devices even though the query complexity is optimal.

Their alternative replaces Grover's diffusion reflection by the transverse-field mixer

$$B = \sum_{j=1}^n X_j,$$

where each X_j is a single-qubit Pauli- X operator. The problem Hamiltonian is taken to be

$$C_u = -|u\rangle\langle u|,$$

for the unknown marked string u . After relabeling the marked item to $u = 0$, we write

$$C \equiv C_0 = -|0\rangle\langle 0|,$$

and the initial state remains the uniform product state

$$|+\rangle^{\otimes n},$$

and the dynamics generated by both B and C preserve the permutation-symmetric subspace of dimension $n+1$ [14].

The algorithm applies oracle and mixer unitaries periodically through the single-period building block

$$W(\gamma) = e^{-i\pi B/n} e^{i\gamma C} e^{-i\pi B/n} e^{-i\gamma C},$$

with $\gamma \in (0, \pi]$ a tunable phase parameter. Jiang et al. show that the choice $\gamma = \pi$ yields the best asymptotic success behavior in their analysis. Although this construction does not use Grover's Euclidean reflection picture directly, it is still governed by a low-dimensional effective transition: using spin-coherent-state and phase-space methods, the paper shows that repeated application of $W(\pi)$ induces a nearly closed transition between states having large overlap with the initial uniform state and the marked computational basis state [14].

From the perspective of this survey, the significance of this result is twofold. First, it shows that the quadratic quantum speedup of unstructured search does not depend on implementing Grover's exact diffusion operator gate by gate. Second, it provides a natural conceptual bridge from circuit-model amplitude amplification to later Hamiltonian- and QAOA-inspired alternating-operator frameworks. The resulting average query complexity is

$$T(n) \simeq \frac{\pi}{2\sqrt{2}} 2^{n/2} = \frac{\pi}{2\sqrt{2}} \sqrt{N},$$

which retains the optimal $\Theta(\sqrt{N})$ scaling while differing from Grover's absolute optimum only by a small constant factor. In exchange, the mixer layer is built entirely from single-qubit transverse-field rotations, which is substantially more favorable for hardware with costly entangling gates.

V. FIXED-POINT AMPLITUDE AMPLIFICATION

A. The Root Cause: Unitarity and Instability

Grover's iterate G is strictly unitary, so its action in the $\text{span}\{| \text{Bad} \rangle, | \text{Good} \rangle\}$ plane is a *continuous rotation*. This creates the “soufflé problem”: the success probability rises strictly to unity and then inevitably falls again. Thus, if the number of marked items M (equivalently the initial success probability $p = M/N$) is unknown, there is no principled stopping time.

Furthermore, as highlighted by the high-density edge case where $p > 1/2$ at the conclusion of Section IV, this continuous rotation can overshoot the target $\mathcal{H}_{\text{Good}}$ subspace even after a single query. Fixed-point amplitude amplification (FPAA) addresses both this sinusoidal instability and the high-density overshooting problem by designing a *sequence* of phase-shifted Grover iterates whose final success probability remains above $1 - \delta^2$ over a broad *range* of marked fractions, with $\delta \in (0, 1)$ a user-chosen error tolerance.

B. Generalized Grover Iterate

Fix the target subspace projector $\Pi_{\text{Good}} = \sum_{x: f(x)=1} |x\rangle\langle x|$ onto $\mathcal{H}_{\text{Good}}$, and the uniform initial state $|\text{All}\rangle$. Let $p = \|\Pi_{\text{Good}} |\text{All}\rangle\|^2$ denote the initial success probability.

Introduce generalized phase reflections about the target and start state:

$$S_{\text{Good}}(\beta) = I - (1 - e^{i\beta}) |\text{Good}\rangle\langle \text{Good}|, \quad (22)$$

$$S_{\text{All}}(\alpha) = I - (1 - e^{-i\alpha}) |\text{All}\rangle\langle \text{All}|. \quad (23)$$

The generalized Grover iterate is then defined strictly as:

$$G(\alpha, \beta) = -S_{\text{All}}(\alpha) S_{\text{Good}}(\beta). \quad (24)$$

Standard Grover search is recovered as the special case where all generalized phases equal π .

C. Success Probability and Chebyshev Mapping

The key design goal in FPAA is to explicitly replace the rigid sinusoidal dependence of standard Grover amplification by a synthesized polynomial map exhibiting a flat high-success “passband.” While standard Grover amplification creates a sharp, oscillating sine wave of success probability, the Chebyshev sequence effectively “flattens” the peak of that wave into a wide, stable plateau. This ensures that once the state vector enters the target region, the variable phase shifts mathematically dampen the rotation, actively preventing it from leaving. Following [15], for an odd integer $L = 2l + 1$ one uses a palindromic

sequence of l generalized iterates

$$S_L = G(\alpha_l, \beta_l) \cdots G(\alpha_2, \beta_2) G(\alpha_1, \beta_1), \quad (25)$$

with query complexity $L - 1 = 2l$. This sequence acts on the two-dimensional invariant subspace so that the final success probability can be written exactly as a polynomial transformation $P_L(p)$ of the initial success probability p .

In the construction of Yoder, Low, and Chuang (2014) [15], one obtains the explicit Chebyshev-form map

$$P_L(p) = 1 - \delta^2 T_L^2(T_{1/L}(1/\delta) \sqrt{1-p}), \quad (26)$$

where T_L is the L -th Chebyshev polynomial of the first kind and $T_{1/L}(\cdot)$ denotes the functional inverse of $T_L(\cdot)$ defined strictly on $[1, \infty)$ (equivalently calculated as $T_{1/L}(x) = \cosh(\frac{1}{L} \text{arcosh } x)$ for $x \geq 1$).

Intuition. Standard Grover amplification produces a sinusoidal success probability curve that can overshoot the optimal point and oscillate indefinitely. Fixed-point amplitude amplification replaces this oscillatory behavior with a Chebyshev-polynomial-shaped response that creates a broad, flat plateau of high success probability. Once p exceeds the passband edge w , the polynomial $P_L(p)$ is bounded from below by $1 - \delta^2$, regardless of the exact value of p . This makes FPAA robust to unknown or variable target densities in a way that standard Grover is not.

1. Fixed-Point Property

We mathematically define the passband edge as the threshold variable w :

$$w = 1 - T_{1/L}(1/\delta)^{-2}. \quad (27)$$

Then for all initial probabilities $p \geq w$, the sequence satisfies

$$P_L(p) \geq 1 - \delta^2. \quad (28)$$

2. Complexity

The exact finite- L fixed-point requirement is

$$1 - T_{1/L}(1/\delta)^{-2} \leq p. \quad (29)$$

For asymptotically small values of p , this yields the sufficient scaling estimate

$$L = \mathcal{O}\left(\frac{\log(2/\delta)}{\sqrt{p}}\right), \quad (30)$$

and the query complexity is $L - 1$. Thus FPAA preserves Grover's optimal quadratic scaling while providing fixed-point robustness against unknown initial densities.

j	α_j	β_j
1	$2 \cot^{-1} \left(\tan(2\pi/3) \sqrt{1-\gamma^2} \right)$	$-2 \cot^{-1} \left(\tan(2\pi/3) \sqrt{1-\gamma^2} \right)$

TABLE IV. Palindromic FPAA sequence phases for $L = 3$ (equivalently, $l = 1$ generalized Grover iterate).

D. Phase Schedule Derivation

The Chebyshev design in (26) can be implemented *exactly* by choosing an analytic phase schedule $\{(\alpha_j, \beta_j)\}_{j=1}^l$ for odd $L = 2l + 1$. Let

$$\gamma^{-1} = T_{1/L}(1/\delta), \quad \text{so that} \quad \gamma \in (0, 1]. \quad (31)$$

The discrete phases are then given by

$$\begin{aligned} \alpha_j &= 2 \cot^{-1} \left(\tan(2\pi j/L) \sqrt{1-\gamma^2} \right), \\ \beta_{l-j+1} &= -\alpha_j, \end{aligned} \quad (32)$$

for $j = 1, \dots, l$, so the schedule is palindromic.

E. Recursive Nesting

An important structural feature of the FPAA family is *recursive nesting*. In the exact Yoder–Low–Chuang construction, if an outer sequence of length L_2 targets final error bound δ , then the inner sequence of length L_1 must be assigned the retuned bound

$$\delta_1 = T_{1/L_2}(1/\delta)^{-1}. \quad (33)$$

With that choice, composing the two fixed-point blocks yields

$$T_{L_2}(T_{L_1}(x)) = T_{L_1 L_2}(x), \quad (34)$$

so one obtains a new fixed-point sequence of effective computational length $L_1 L_2$ without restarting from the fiducial $|\text{All}\rangle$ state. Operationally, the state output of the first FPAA block becomes the input to the next block, and the polynomial degree multiplies under composition. This recursive architecture provides a natural way to broaden the target passband (or iteratively reduce δ) without incurring restart overhead.

F. Phase Table for $L = 3$

For structural and practical implementation, it is mathematically convenient to tabulate the palindromic phase schedule derived from (32) for small odd values of L . For exactly $L = 3$, one has $l = 1$ and $\gamma^{-1} = T_{1/3}(1/\delta)$, so

$$\alpha_1 = 2 \cot^{-1} \left(\tan(2\pi/3) \sqrt{1-\gamma^2} \right), \quad \beta_1 = -\alpha_1. \quad (35)$$

Then the exact fixed-point schedule evaluates to:

G. From Geometry to Algebraic Synthesis

The structural formulation of Fixed-Point Amplitude Amplification marks an important shift in perspective: it moves from *pure geometry* (rigid spatial rotations defined by fixed vectors) toward *algebraic synthesis* (designing operator sequences to approximate bounded classical polynomials). The idea of interleaving generalized target reflections with state reflections to shape a transformation polynomial provides a useful conceptual precursor to Quantum Singular Value Transformation (QSVT), which we discuss in Section XI.

VI. OBLIVIOUS AMPLITUDE AMPLIFICATION

A. The Core Problem: The Bottlenecks of “Oblivious” Amplification

Section V removed the stopping-time instability of Grover rotations when p is unknown. However, both Grover and FPAA still assume the ability to reflect about a known start state $|\text{All}\rangle$. Oblivious amplification removes that assumption and studies the regime where the input is unknown [12].

In this setting, the input is an arbitrary pure state $|\psi\rangle$ rather than a known basis-derived initialization. The key question is whether amplitude amplification can still be implemented without explicit knowledge of $|\psi\rangle$. This subsection formalizes the obstruction and then identifies the additional structure needed for oblivious amplification to become possible.

B. Generalized Algebraic Setup

Let $|\psi\rangle$ be an arbitrary, unknown n -qubit pure state. We are given black-box access to a Boolean predicate $f : \{0, 1\}^n \rightarrow \{0, 1\}$ representing the target index, and an initialization unitary $\mathcal{A}$ such that $\mathcal{A}|\psi\rangle = \sum_x \alpha_x |x\rangle$.

Define the success probability analytically as $p(\psi) = \sum_{f(x)=1} |\alpha_x|^2$. We parameterize the normalized good and bad mathematical components discretely as:

$$\begin{aligned} |\text{Good}(\psi)\rangle &= \frac{1}{\sqrt{p(\psi)}} \sum_{f(x)=1} \alpha_x |x\rangle, \\ |\text{Bad}(\psi)\rangle &= \frac{1}{\sqrt{1-p(\psi)}} \sum_{f(x)=0} \alpha_x |x\rangle. \end{aligned} \quad (36)$$

By direct superposition,

$$\mathcal{A}|\psi\rangle = \sqrt{1-p(\psi)} |\text{Bad}(\psi)\rangle + \sqrt{p(\psi)} |\text{Good}(\psi)\rangle.$$

Unlike the standard setting, both $p(\psi)$ and the associated two-dimensional amplification plane depend on the unknown input. The rotation subspace is therefore not known a priori.

C. Roadblocks: State Collapse and Reflection Obstruction

In standard amplification, failed runs can be discarded and restarted from $|0\rangle^{\otimes n}$. In the oblivious setting, one typically has only coherent access to $|\psi\rangle$. Measuring a success flag collapses the state into either $|\text{Good}(\psi)\rangle$ or $|\text{Bad}(\psi)\rangle$, and no-cloning prevents generic backup copies. Hence the usual repeat-until-success strategy is unavailable.

The deeper unmitigable obstruction remains strictly geometric. In standard implementations, the diffusion operator is constructed systematically as $R = \mathcal{A}(I - 2|0\rangle^{\otimes n}\langle 0|^{\otimes n})\mathcal{A}^{-1}$. This structure relies precisely on $|0\rangle^{\otimes n}$ operating as a globally known, fixed basis state whose geometric reflection is efficiently algorithmically synthesizable.

In the oblivious model, the corresponding reflection would be $R_{|\psi\rangle} = I - 2|\psi\rangle\langle\psi|$. Implementing this for unknown $|\psi\rangle$ is not generally possible without state information, and a universal construction would conflict with linearity across non-orthogonal inputs. Therefore the geometric Grover-style mechanism cannot be transplanted directly.

Even if one imposes $p(\psi) \equiv p$ over the relevant state family, generic oblivious amplification still requires state-dependent reflections. The nontrivial case is therefore possible only under additional structure (for example, purified ancilla layout / block-encoding structure), which we now formalize.

D. The Purified Setting and Ancilla Padding

General oblivious amplification fails because both restart and unknown-state reflections are unavailable. The standard workaround is a stricter model: the *purified setting with independent initial weight*.

Assume a tensor split $\mathcal{H} = \mathcal{H}_{\text{ancilla}} \otimes \mathcal{H}_{\text{data}}$ with dimensions 2^l and 2^m ($n = l + m$), and promise inputs of the form

$$|\psi\rangle = |0\rangle^{\otimes l} |\phi\rangle,$$

where $|\phi\rangle$ is arbitrary but the ancilla prefix is known and clean. The requirement $l > 0$ provides the fixed reference subspace needed for oblivious constructions and is precisely the structural ingredient used in block-encoding pipelines.

The purified setting identically imposes a singular, crucial invariance constraint onto the success probability evaluated after applying the global preparation unitary $\mathcal{A}$:

$$p = \left\| \Pi_{\text{Good}} \mathcal{A} |0\rangle^{\otimes l} |\phi\rangle \right\|_2^2. \quad (37)$$

Specifically, p must be independent of $|\phi\rangle$ on the relevant data subspace, i.e., $p(\phi) \equiv p$. This invariance removes

the input-dependent drift that obstructed the geometric amplification picture above.

E. The Identity Block Derivation

To structurally decode this condition, define the Hermitian operator $M = \mathcal{A}^\dagger \Pi_{\text{Good}} \mathcal{A}$. Evaluating the target probability formally yields $p = \langle 0^{\otimes l} | \langle \phi | M | 0^{\otimes l} \rangle | \phi \rangle$. By rigidly constraining the auxiliary portion of the evaluation to the zero-basis $|0\rangle^{\otimes l}$, this expression cleanly isolates a specific principal matrix block of the global operator M . Indexing the complete computational basis geometrically, we structurally partition M into targeted functional blocks:

$$M = \begin{bmatrix} M_{\text{TL}} & M_{\text{TR}} \\ M_{\text{BL}} & M_{\text{BR}} \end{bmatrix}, \quad (38)$$

where M_{TL} constitutes all computational dimensions governed strictly by the $0^{\otimes l}$ auxiliary prefix. The initial state $|0\rangle^{\otimes l} |\phi\rangle$ is linearly bounded completely within this single operational quadrant, fundamentally collapsing the geometric evaluation to $p = \langle \phi | M_{\text{TL}} | \phi \rangle$.

Because this expectation must be constant for all pure $|\phi\rangle$, spectral decomposition forces every eigenvalue of M_{TL} to equal p .

This deterministic condition analytically validates the decisive overarching structural equality:

$$M_{\text{TL}} = pI. \quad (39)$$

Thus, on the clean-ancilla sector, $\mathcal{A}^\dagger \Pi_{\text{Good}} \mathcal{A}$ acts as a scalar multiple of the identity. In the degenerate case $l = 0$, this collapses to the global relation $M = pI$, recovering the trivial boundary where nontrivial oblivious amplification is unavailable.

F. Synthesizing the Oblivious Reflection

The identity block $M_{\text{TL}} = pI$ replaces the unavailable unknown-state reflection. Let

$$C = |0\rangle^{\otimes l} \langle 0|^{\otimes l} \otimes I_{\text{data}}.$$

From $\mathcal{A}^{-1} \Pi_{\text{Good}} \mathcal{A} = M$, projection gives

$$CM |0\rangle^{\otimes l} |\phi\rangle = p |0\rangle^{\otimes l} |\phi\rangle,$$

uniformly over $|\phi\rangle$. Therefore the reflection about the clean-ancilla sector is

$$R_{\text{clean}} = 2C - I,$$

which plays the role needed inside the effective two-dimensional amplification subspace. The resulting oblivious iterate

$$Q = \mathcal{A}(2C - I)\mathcal{A}^{-1}(I - 2\Pi_{\text{Good}})$$

recovers Grover-type rotation with $\mathcal{O}(1/\sqrt{p})$ scaling, without tomography, cloning, or measurement-driven restarts.

G. The Equivalence to Unitary Block Encodings

The purified setting above is equivalent to the block-encoding viewpoint used in modern quantum algorithms (including Hamiltonian simulation, linear algebra primitives, and QSVT).

A defined unitary operator $\mathcal{A}$ formally evaluates as a block-encoding modeling a generic system operator B if its internal spatial matrix rigidly coordinates strictly as:

$$\mathcal{A} = \begin{bmatrix} B & * \\ * & * \end{bmatrix}, \quad (40)$$

Restricting to the clean ancilla therefore isolates the effective transformation B .

If $\mathcal{A}$ block-encodes $B = \sqrt{p}U$, then for any $|\phi\rangle$,

$$\mathcal{A}|0\rangle^{\otimes l}|\phi\rangle = \sqrt{p}|0\rangle^{\otimes l}U|\phi\rangle + |\perp(\phi)\rangle,$$

so the success amplitude in the designated ancilla sector is $\sqrt{p}$ and the corresponding success probability is p , independent of $|\phi\rangle$. So block encoding implies the purified invariance condition.

Conversely, specialize the success predicate to the clean-ancilla sector itself, so that $\Pi_{\text{Good}} = C$, exactly as in the block-encoding proof. Then

$$M = \mathcal{A}^\dagger C \mathcal{A},$$

and its clean-ancilla block is

$$M_{\text{TL}} = \mathcal{A}_{\text{TL}}^\dagger \mathcal{A}_{\text{TL}}.$$

Since the purified condition yields $M_{\text{TL}} = pI$, it follows that $\mathcal{A}_{\text{TL}}^\dagger \mathcal{A}_{\text{TL}} = pI$, and therefore $\mathcal{A}_{\text{TL}} = \sqrt{p}U$ for some unitary U . Hence the two formalisms are equivalent.

Final Verdict. The fundamental relational dynamic proving the equivalence:

$$\boxed{\begin{array}{c} \text{Purified Amplification Setting} \\ \Longleftrightarrow \\ \text{Unitary Block Encoding} \end{array}} \quad (41)$$

establishes OAA as the coherent amplification primitive for subnormalized block encodings, and prepares the transition to Section XI.

VII. CONTROLLED QUANTUM AMPLIFICATION

A. From Detection to Finding

Standard Grover amplification is designed so that the amplified state itself has large overlap with a marked basis state. In quantum-walk search, however, one often begins with a search operator

$$A = WG,$$

where W is a walk/update unitary whose $(+1)$ -eigenvector is the raw starting state $|\text{init}_{\text{raw}}\rangle$, and

$$G = I - 2|g\rangle\langle g|$$

is the reflection about a target state $|g\rangle$ [13]. Spectral information about A may suffice to *detect* that a target exists, but it does not automatically imply that repeated use of A will *produce* $|g\rangle$ with constant probability upon measurement.

Controlled Quantum Amplification (CQAA) resolves this detection-versus-finding bottleneck by wrapping the search dynamics in a one-qubit control register. The resulting controlled circuit converts abstract detection procedures into actual finding procedures while preserving the same asymptotic quantum hitting time [13].

B. Controlled Architecture and the Principal Eigenvector

Dohotaru and Høyer introduce a control qubit together with the rotated basis

$$\begin{aligned} |\tilde{0}\rangle &= \cos(\tilde{\theta})|0\rangle + \sin(\tilde{\theta})|1\rangle, \\ |\tilde{1}\rangle &= -\sin(\tilde{\theta})|0\rangle + \cos(\tilde{\theta})|1\rangle. \end{aligned}$$

The controlled search circuit is built from two conditional operators,

$$|\tilde{0}\rangle\langle\tilde{0}| \otimes G + |\tilde{1}\rangle\langle\tilde{1}| \otimes I$$

and

$$|0\rangle\langle 0| \otimes W + |1\rangle\langle 1| \otimes I,$$

whose product we denote by U .

Let

$$\sin(\theta) = |\langle g | \text{init}_{\text{raw}} \rangle|, \quad \epsilon = \sin^2 \theta,$$

and define the orthogonalized search-plane start vector

$$|\text{init}_\perp\rangle = \frac{|\text{init}_{\text{raw}}\rangle - \sin(\theta)|g\rangle}{\cos(\theta)}.$$

In the sparse-target regime $\epsilon < 1/2$, one may choose $\tilde{\theta}$ so that

$$\sin(\tilde{\theta}) = \frac{\sin(\theta)}{\cos(\theta)}.$$

With this tuning, the circuit acquires a distinguished $(+1)$ -eigenvector whose normalized form is

$$|U_0\rangle = \frac{1}{\sqrt{2}} \left(|0, \text{init}_\perp\rangle - |\tilde{1}, g\rangle \right).$$

This is the key structural step: the principal eigenspace of U now simultaneously contains the initial configuration and the marked configuration with constant overlap. In this way, the controlled wrapper converts an abstract walk/detection process into a finding process on the enlarged Hilbert space.

C. Quantum Hitting Time Formulation

The paper then redefines quantum hitting time directly in terms of the nontrivial eigenphases of a real unitary. If

$$|w\rangle = w_0 |U_0\rangle + \sum_{j=1}^m w_j (|U_j^+\rangle + |U_j^-\rangle) + w_{-1} |U_{-1}\rangle,$$

where the pairs $|U_j^\pm\rangle$ have eigenvalues $e^{\pm i\alpha_j}$ with $0 < \alpha_j < \pi$, then Dohotaru and Høyer define

$$\text{QHT}_\alpha(U, |w\rangle) = \sqrt{2 \sum_{j=1}^m |w_j|^2 \frac{1}{\alpha_j^2}}.$$

They prove two linked statements. First, there is an algorithm using an expected number of U applications of order

$$\text{QHT}(U, |0, \text{init}_\perp\rangle)$$

that outputs a state with constant overlap on $|\tilde{1}, g\rangle$. Second, the controlled circuit inherits the same asymptotic hitting time as the original search operator:

$$\text{QHT}(U, |0, \text{init}_\perp\rangle) = \Theta(\text{QHT}(A, |\text{init}_\perp\rangle)).$$

Together, these results establish the paper's central message: in this framework, *finding is asymptotically as easy as detecting*.

D. Random-Walk Consequences and Interpolated Walks

For reversible random walks with a unique marked element, the controlled construction yields a quantum algorithm with cost

$$\tilde{\mathcal{O}}\left(S + \sqrt{H}U + \frac{1}{\sqrt{\epsilon}}C\right),$$

where S is setup cost, U is update cost, C is checking cost, and H is the classical hitting time [13]. The same analysis also produces a classical algorithm of cost

$$\mathcal{O}\left(S + HU + \frac{1}{\epsilon}C\right),$$

which the authors derive by quantum arguments.

Another important contribution for the purposes of this survey is the relation to interpolated walks. Rather than explicitly running a quantum interpolated walk $W(s)$, the paper constructs an inner-product-preserving embedding E_s such that

$$E_s W(s) = U(\tilde{\theta}) E_s.$$

Thus controlled quantum amplification can simulate interpolated-walk dynamics directly inside the controlled-amplification framework. This makes CQAA a natural conceptual bridge between Grover-style reflections and the more general walk-based amplification methods that sit adjacent to modern singular-value and block-encoding techniques.

VIII. FIXED-POINT OBLIVIOUS AMPLITUDE AMPLIFICATION

As established in Section VI, OAA removes dependence on known-input reflections. But it still inherits Grover-style overshoot when the subnormalization (equivalently, effective success probability p) is unknown or drifting [16]. Fixed-Point Oblivious Quantum Amplitude Amplification (FOQA) addresses this by introducing a damping schedule that enforces monotonic convergence.

A. Duality Computing and the Dampened Ancestral Projector

Because unknown-state reflections remain unavailable, FOQA cannot directly reuse the Section V schedules on the raw state. Instead it is implemented through a Linear Combination of Unitaries (LCU) construction.

As explicitly derived across Section VI, the unnormalized initialization sequence operates strictly as:

$$A|0\rangle^{\otimes l}|\phi\rangle = \sqrt{p}|\text{Good}(\phi)\rangle + \sqrt{1-p}|\text{Bad}(\phi)\rangle. \quad (42)$$

Instead of a fixed reflection angle about the clean-ancilla projector $C = |0\rangle^{\otimes l}\langle 0|^{\otimes l} \otimes I_{\text{data}}$, FOQA uses a variable damping angle α_n applied conditionally through the LCU step. This yields controlled, non-oscillatory convergence without measuring or learning $|\phi\rangle$.

B. Damped Iteration Mathematics

At iteration n , FOQA defines the wave-division damping operator logically constrained by α_n :

$$V_n = \begin{pmatrix} \cos \frac{\alpha_n}{2} & -\sin \frac{\alpha_n}{2} \\ \sin \frac{\alpha_n}{2} & \cos \frac{\alpha_n}{2} \end{pmatrix}. \quad (43)$$

Although the primitive wave-division gate V_n itself carries the half-angle $\alpha_n/2$, the full five-step LCU iterate analyzed in [16] produces the following full-angle recurrence. After the n -th unsuccessful iteration the working register is parameterized by amplitudes t_n and s_n on the target and non-target branches, respectively, with $|t_n|^2 + |s_n|^2 = 1$.

The per-iteration success probability governing algorithmic termination at step $n+1$ is

$$p_{n+1} = \sin^2(\alpha_n) |t_n|^2. \quad (44)$$

Conditioned on continuation, the target amplitude obeys:

$$t_{n+1} = \frac{t_n \cos(\alpha_n) \cos(2\theta) + s_n \sin(2\theta)}{\sqrt{1 - p_{n+1}}}, \quad (45)$$

and the non-target amplitude satisfies

$$s_{n+1} = \frac{-t_n \cos(\alpha_n) \sin(2\theta) + s_n \cos(2\theta)}{\sqrt{1 - p_{n+1}}}, \quad (46)$$

where $\sin(\theta) = \sqrt{p}$. The overall failure probability after n iterations is therefore

$$q_n = \prod_{k=1}^n (1 - p_k). \quad (47)$$

C. Choice of Damping Schedule

The key design choice is the schedule for α_n :

- **Small** α_n : close to oscillatory Grover behavior.
- **Large** α_n : overdamped behavior.
- **Intermediate** schedules: critical damping with stable monotone gain.

Because $q_n = \prod_{k=1}^n (1 - p_k)$ with $0 \leq p_k \leq 1$, the sequence $\{q_n\}$ is nonincreasing. The schedule $\{\alpha_n\}$ is chosen so that the success probability increases without the Grover-style overshoot associated with fixed-angle iterations.

D. Convergence Guarantee

The resulting guarantees are summarized below.

Theorem 1 (Fixed-Point Oblivious Amplification Limit). *Let $\mathcal{A}$ prepare*

$$\mathcal{A}|0\rangle^{\otimes l}|\phi\rangle = \sqrt{p}|\text{Good}(\phi)\rangle + \sqrt{1-p}|\text{Bad}(\phi)\rangle, \quad (48)$$

with unknown success probability $p \in (0, 1)$.

Then, in the fixed-point damping regime described in [16], FOQA ensures:

1. **Monotone failure reduction:** the continuation probability q_n is nonincreasing.
2. **Fixed-point behavior:** $q_n \rightarrow 0$ as $n \rightarrow \infty$ under the prescribed schedule.
3. **Average complexity:** the mean number of iterations remains $\mathcal{O}(1/\sqrt{p})$, up to constant factors.
4. **No oscillatory overshoot:** the damping schedule suppresses the over-rotation behavior present in fixed-angle Grover iterations.

Therefore FOQA preserves $\mathcal{O}(1/\sqrt{p})$ scaling up to constant factors while eliminating oscillatory instability in the oblivious setting.

E. Architectural Significance

FOQA has three structural consequences:

1. It provides fixed-point monotonic convergence without known-state reflections.

2. It preserves near-optimal quadratic query behavior.
3. It composes naturally with LCU/block-encoding based algorithmic pipelines.

This closes the geometric-to-algebraic bridge from oblivious amplification to the operator-transform viewpoint used in Section XI.

IX. DISTRIBUTED AMPLITUDE AMPLIFICATION

A. Motivation: NISQ Hardware Constraints

The amplitude amplification procedures discussed in the previous sections assume the availability of a single coherent quantum processor capable of executing circuits on an n -qubit Hilbert space of dimension 2^n . While theoretically elegant, this assumption becomes increasingly unrealistic on present-day Noisy Intermediate-Scale Quantum (NISQ) hardware. Current devices suffer from severe constraints on qubit counts, limited coherence times, calibration drift, and connectivity restrictions. This motivates distributed amplitude amplification architectures as developed recently in [17].

Distributed Quantum Amplitude Amplification (DQAA) addresses these architectural limitations by partitioning the global search space across multiple smaller quantum processors. Rather than executing the search algorithm on a monolithic n -qubit device, the algorithm distributes subproblems across 2^j processors, each operating on only $n - j$ qubits. The processors execute amplitude amplification locally, and the classical controller subsequently aggregates the measurement outcomes.

It is crucial to emphasize that this approach does not alter the fundamental asymptotic complexity of quantum search. The optimal query complexity $\mathcal{O}(\sqrt{N/M})$ established for Grover's algorithm remains unchanged. Instead, DQAA should be interpreted as an *architectural redistribution* of the computation, trading hardware width for reduced per-device qubit requirements.

B. Partitioning the Global Search Space

Let the global search function be

$$f : \{0, 1\}^n \rightarrow \{0, 1\},$$

where $f(x) = 1$ indicates a marked solution.

We partition the input register into a prefix of j qubits and a suffix of $n - j$ qubits,

$$x = (i_k, y), \quad i_k \in \{0, 1\}^j, \quad y \in \{0, 1\}^{n-j}.$$

The prefix i_k indexes the processor responsible for evaluating the corresponding subspace. Each processor therefore receives a restricted oracle

$$f_k(y) = f(i_k y).$$

Thus the global function decomposes into 2^j independent sub-functions

$$f_k : \{0, 1\}^{n-j} \rightarrow \{0, 1\}.$$

Each processor performs amplitude amplification within the Hilbert space

$$\mathcal{H}_k = \text{span}\{|y\rangle : y \in \{0, 1\}^{n-j}\}.$$

C. Tensor Decomposition of the Initialization Operator

A key structural requirement for distributed amplification is that the global initialization unitary admits a strict tensor decomposition

$$\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2,$$

where $\mathcal{A}_1$ acts on the prefix register and $\mathcal{A}_2$ acts on the suffix register. The global initial state therefore factorizes as

$$\mathcal{A}|0\rangle^{\otimes n} = (\mathcal{A}_1|0\rangle^{\otimes j}) \otimes (\mathcal{A}_2|0\rangle^{\otimes(n-j)}).$$

Within processor k , the local suffix state is therefore

$$|\psi_k\rangle = \mathcal{A}_2|0\rangle^{\otimes(n-j)}.$$

Using the decomposition introduced in Section IV, the local Hilbert space separates into

$$\mathcal{H}_k = \mathcal{H}_{\text{Good},k} \oplus \mathcal{H}_{\text{Bad},k}.$$

Let

$$p_k = \langle \psi_k | \Pi_{\text{Good},k} | \psi_k \rangle$$

denote the local success probability. Equivalently, one may write

$$|\psi_k\rangle = \sqrt{p_k} |\text{Good}_k\rangle + \sqrt{1-p_k} |\text{Bad}_k\rangle,$$

where $|\text{Good}_k\rangle \in \mathcal{H}_{\text{Good},k}$ and $|\text{Bad}_k\rangle \in \mathcal{H}_{\text{Bad},k}$ are normalized and orthogonal.

a. Limitations of the Tensor Assumption. The requirement $\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2$ is highly restrictive. In many modern quantum algorithms the initialization procedure generates significant entanglement across registers. In particular, block-encoded operators and Quantum Singular Value Transformation (QSVT) pipelines rely on entangled system-ancilla registers that generally do not admit simple tensor factorizations. Consequently, distributed amplitude amplification is primarily applicable to algorithms whose initial states are separable across the partition boundary.

D. Local Probability Obstruction

Suppose the global algorithm has success probability

$$p = \sum_{x: f(x)=1} |\alpha_x|^2.$$

Although this quantity may be known or bounded, its distribution across the 2^j partitions is generally unknown. The local success probabilities p_k may vary dramatically between nodes. Consequently, the standard Grover stopping rule cannot be applied locally because the required iteration count depends on p_k .

The following theorem addresses this obstruction.

Theorem 2 (Lucky Node Theorem). *Let p denote the global success probability and p_k the success probability associated with partition k . Then there exists at least one node $\hat{k}$ such that*

$$p_{\hat{k}} \geq p.$$

Proof. Using $\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2$, write the initialized global state as

$$|\Psi\rangle = \sum_{k=0}^{2^j-1} \sum_{y \in \{0,1\}^{n-j}} \alpha_{k,y} |i_k\rangle |y\rangle.$$

For each partition k , define

$$a_k = \sum_{y: f_k(y)=1} |\alpha_{k,y}|^2, \quad b_k = \sum_{y: f_k(y)=0} |\alpha_{k,y}|^2,$$

so $w_k := a_k + b_k$ is the total probability weight of partition k and $\sum_k w_k = 1$. If $w_k > 0$, define local success probability

$$p_k = \frac{a_k}{w_k}, \quad 0 \leq p_k \leq 1.$$

The global success probability is

$$p = \sum_{k=0}^{2^j-1} a_k = \sum_{k=0}^{2^j-1} w_k p_k.$$

Assume for contradiction that $p_k < p$ for all k with $w_k > 0$. Multiplying by $w_k \geq 0$ and summing gives

$$\sum_k w_k p_k < \sum_k w_k p = p \sum_k w_k = p.$$

This implies $p < p$, a contradiction. Therefore there exists at least one $\hat{k}$ such that $p_{\hat{k}} \geq p$. $\square$

This theorem shows that there exists a *lucky node* whose initial success probability is at least as large as the global value.

E. Distributed Fixed-Point Amplification

Because the Lucky Node Theorem implies that $p_k \geq p$ for at least one node, the Fixed-Point Amplitude Amplification (FPAA) procedure of Section V can be applied at each processor using the global lower bound p .

Each processor executes the local iteration operator

$$Q_k = -\mathcal{A}_2 S_0 \mathcal{A}_2^\dagger S_{f_k},$$

where

$$S_{f_k} = I - 2\Pi_{\text{Good},k}$$

is the phase oracle for the local function f_k , and

$$S_0 = I - 2|0\rangle^{\otimes(n-j)}\langle 0|^{\otimes(n-j)}.$$

This is the reflection about the all-zero state of the suffix register.

The phase sequence used in these reflections follows the Chebyshev fixed-point schedule derived previously in Section V. Each processor therefore executes the same FPAA sequence optimized for the lower bound p .

F. Global Success Guarantee

Theorem 3. *Let each processor execute the FPAA schedule with l generalized iterates satisfying*

$$l \geq \frac{1}{2\sqrt{p}} \log\left(\frac{2}{\epsilon}\right).$$

Equivalently, the corresponding odd sequence length satisfies

$$L = 2l + 1 = \mathcal{O}\left(\frac{1}{\sqrt{p}} \log \frac{2}{\epsilon}\right).$$

Then the distributed algorithm succeeds with probability at least

$$1 - \epsilon^2.$$

Proof. By the Lucky Node Theorem there exists a node $\hat{k}$ with $p_{\hat{k}} \geq p$. Running FPAA with

$$l \geq \frac{1}{2\sqrt{p}} \log\left(\frac{2}{\epsilon}\right)$$

therefore also satisfies the amplification requirement for node $\hat{k}$, so its final success probability obeys

$$P_{\hat{k}} \geq 1 - \epsilon^2.$$

Let P_k denote the final success probability at node k . Since node executions are independent, the global success probability is

$$P_{\text{tot}} = 1 - \prod_{k=0}^{2^j-1} (1 - P_k).$$

Hence

$$P_{\text{tot}} \geq P_{\hat{k}} \geq 1 - \epsilon^2.$$

Therefore the distributed algorithm succeeds with probability at least $1 - \epsilon^2$. $\square$

G. Complexity Analysis

The parallel runtime of the distributed algorithm is

$$\mathcal{O}\left(\frac{1}{\sqrt{p}} \log\left(\frac{2}{\epsilon}\right)\right),$$

which matches fixed-point amplitude amplification up to the same logarithmic error factor.

However, the aggregate computational work across all processors is

$$2^j \cdot \mathcal{O}\left(\frac{1}{\sqrt{p}} \log\left(\frac{2}{\epsilon}\right)\right).$$

Thus DQAA preserves Grover optimality while trading increased hardware parallelism for reduced per-device circuit size.

H. Empirical Validation

To validate the theoretical guarantees in a concrete setting, we performed a numerical simulation on a small distributed search instance.

The global problem used an $n = 6$ qubit search space ($N = 64$). We chose partition parameter $j = 2$, yielding $2^j = 4$ independent processors, each operating on a suffix register of $n - j = 4$ qubits. Each processor therefore ran fixed-point amplitude amplification over a local space of dimension $2^4 = 16$.

Two distinct behaviors were observed:

- **Unlucky nodes** ($p_k = 0$): if a partition contained no marked element, fixed-point iteration could not amplify absent target amplitude. These nodes produced near-uniform measurement outcomes over the local suffix basis, i.e., random-noise-like behavior.
- **Lucky nodes** ($p_k > 0$): partitions containing marked states converged monotonically toward the target subspace. The success probability increased without Grover-style overshoot and exceeded the bound $1 - \epsilon^2$ in the tested runs.

This experiment supports the practical content of the Lucky Node Theorem: under uneven partition statistics, at least one processor can still be amplified reliably to high success probability.

I. Hardware Scaling

Let n denote the global qubit count and j the partition parameter.

Quantity	Value
Processors	2^j
Qubits per processor	$n - j$
Total qubits	$2^j(n - j)$

Three scaling regimes are particularly relevant:

- $j = \mathcal{O}(1)$: modest hardware parallelism with minimal overhead.
- $j = \mathcal{O}(\log n)$: polynomial processor growth.
- $j = \Theta(n)$: exponential processor requirements.

Practical implementations therefore restrict j to small values.

J. Architectural Advantages

Distributed amplification possesses several practical advantages:

- **Reduced qubit requirements.** Each processor requires only $n - j$ qubits.
- **Parallel execution.** All processors operate simultaneously.
- **Zero quantum communication.** Nodes operate independently and communicate only classically after measurement by concatenating local suffix outcomes with their corresponding prefixes.

These properties make DQAA particularly attractive for heterogeneous quantum networks composed of multiple small NISQ devices.

K. Limitations and Relation to QSVT

Despite its architectural advantages, DQAA exhibits important limitations. The strict tensor decomposition requirement $\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2$ prevents its direct application to algorithms whose initialization circuits generate significant entanglement across registers.

This restriction is substantial in modern algorithmic settings. Many ansatz-based state preparations, block-encoded operators, and Quantum Singular Value Transformation (QSVT) pipelines rely on correlated system-ancilla structure rather than separable initialization. In such settings, the partition boundary generally cuts across entangling operations, so the local processors no longer inherit an autonomous amplification problem of the form assumed here. Consequently, extending distributed architectures to QSVT-based or block-encoding-based algorithms remains an open research direction rather than a direct corollary of the present framework.

This limitation marks the boundary between the geometric amplitude amplification techniques developed in the preceding sections and the algebraic operator transformations that characterize modern quantum algorithm design.

L. Distributed Exact Quantum Amplitude Amplification (DEQAAA)

The 2025 DQAA framework shows that amplitude amplification can be parallelized across multiple nodes, but it does so under three significant restrictions: the state-preparation unitary must factor as $\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2$, the number of nodes is constrained to 2^j , and the resulting procedure remains approximate rather than exact. Zhou et al. (2026) [18] address these three bottlenecks simultaneously by proposing a Distributed Exact Quantum Amplitude Amplification Algorithm (DEQAAA) for quantum states with arbitrary amplitude distributions.

Historically, the paper presents DEQAAA as the next step in a distributed-search lineage that runs from earlier distributed Grover-style constructions through Hua and Qiu’s 2025 distributed amplitude amplification framework. In that progression, the 2025 result established that amplification could be spread across many small processors, while the 2026 result replaces the earlier fixed-point, tensor-factorized architecture by a distributed exact-amplification architecture built for arbitrary amplitude distributions [18].

The architectural change is substantial. Instead of requiring a power-of-two node count, DEQAAA supports any number of computing nodes t satisfying $2 \leq t \leq n$. If node j is assigned n_j qubits with $\sum_{j=0}^{t-1} n_j = n$, then the largest local memory requirement is

$$\max(n_0, n_1, \dots, n_{t-1}),$$

rather than the $2^j(n - j)$ total-qubit overhead of the 2025 construction. The comparison table in [18] further

emphasizes that DEQAAA removes the stringent tensor-product assumption on $\mathcal{A}$ and supports exact multi-target amplification in settings where DQAA does not.

The algorithm proceeds in two phases. First, the global target set X_g is partitioned into local target sets by the substring map

$$h(x, s, l) = x_s x_{s+1} \cdots x_{s+l-1},$$

which induces local Boolean subfunctions f_j and node-wise success probabilities p_j [18]. Each node then performs a local exact amplitude amplification step using

$$EQ_j = \mathcal{A}_{|\phi_j\rangle} R_{|0\rangle^{\otimes n_j}}^{\phi_j} \mathcal{A}_{|\phi_j\rangle}^\dagger R_{f_j}^{\phi_j},$$

repeated

$$J_j + 1$$

times, where

$$J_j = \left\lfloor \frac{\pi}{4 \arcsin(\sqrt{p_j})} - \frac{1}{2} \right\rfloor$$

and the phase is

$$\phi_j = 2 \arcsin \left(\frac{\sin(\pi/(4J_j + 6))}{\sqrt{p_j}} \right).$$

If the resulting global success probability after this local phase already satisfies $p'_g = 1$, the algorithm terminates. Otherwise, the full first-phase unitary is bundled into a composite operator

$$\mathcal{B} = \left[\bigotimes_{j=0}^{t-1} (EQ_j^{J_j+1}) \right] \mathcal{A},$$

and a second, global exact amplification step is applied:

$$\widehat{EQ} = \mathcal{B} R_{|0\rangle^{\otimes n}}^{\hat{\phi}} \mathcal{B}^\dagger R_f^{\hat{\phi}}, \quad \hat{J} = \left\lfloor \frac{\pi}{4 \arcsin(\sqrt{p'_g})} - \frac{1}{2} \right\rfloor,$$

with

$$\hat{\phi} = 2 \arcsin \left(\frac{\sin(\pi/(4\hat{J} + 6))}{\sqrt{p'_g}} \right).$$

and the global exact operator is applied $\hat{J} + 1$ times. This two-phase construction is the mechanism by which the paper proves exact final amplification for the global target set [18].

It is also worth emphasizing that the algorithm is not merely “distributed Grover” with a corrected stopping rule. Structurally, both phases are distributed instances of Exact Quantum Amplitude Amplification (EQAAA): the first phase executes local exact amplification on each node, while the second phase applies a global exact-amplification step using $\mathcal{B}$ as the effective state-preparation unitary.

In that sense, DEQAAA should be read as a distributed realization of EQAAA rather than as a fixed-point search method. This marks an important contrast with the 2025 DQAA construction, whose distributed correctness relied on Yoder–Low–Chuang fixed-point schedules precisely because the local success amplitudes were not known in advance.

From the perspective of this survey, DEQAAA is especially notable because it shifts distributed amplification away from the lucky-node logic of the 2025 model and toward explicit phase synthesis. Standard distributed Grover-style constructions depend on rigid sign flips; by contrast, DEQAAA replaces those reflections with parameterized phase-rotation operators R_f^ϕ and $R_{|0\rangle^{\otimes n}}^\phi$, bringing the distributed setting conceptually closer to the phase-engineering viewpoint that culminates in QSVT.

The cost model in [18] is also different from the query-centric analysis used earlier in this survey. The paper studies decomposed circuit depth and proves

$$\text{dep(DEQAAA)} = \begin{cases} \text{dep(First phase)}, & p'_g = 1, \\ \text{dep(First phase)} & \text{otherwise.} \\ + \text{dep(Second phase)}, \end{cases} \quad (49)$$

$$\begin{aligned} \text{dep(First phase)} &= \text{dep}(\mathcal{A}) \\ &+ \max_j \left\{ \text{dep}(j\text{-th node}, p_j, |X_j|) \right\}. \end{aligned} \quad (50)$$

This is paired with an explicit decomposition of the multi-controlled phase-shift gate

$$C^{n-1}PS(\phi),$$

into elementary gates. In the paper’s MindSpore Quantum experiments, the resulting distributed circuits achieve more than a 97% reduction in both gate count and circuit depth in the 10-qubit example relative to centralized QAAA and EQAAA implementations [18].

At the same time, DEQAAA introduces a different hidden cost. The local operators $\mathcal{A}_{|\phi_j\rangle}$ must prepare substates derived from the global amplitude distribution, so the burden of tensor-factorizing $\mathcal{A}$ is replaced by the burden of synthesizing local substates. Moreover, the paper explicitly notes that exact amplification presumes access to the exact probability distribution P of the input state. In realistic experiments one instead estimates an approximate distribution $\tilde{P}$ from repeated measurements and quantifies the mismatch via the Kullback–Leibler divergence

$$\begin{aligned} D_{\text{KL}}(\tilde{P} \| P) &= \sum_x \tilde{P}(x) \\ &\times \log \frac{\tilde{P}(x)}{P(x)}. \end{aligned}$$

The mapping from a local probability distribution P_j to a substate $|\phi_j\rangle$ is also not unique; the paper explicitly allows any substate consistent with P_j and points to

Feature	DQAA (2025)	DEQAAA (2026)
Exact solution	No	Yes
Stringent assumption on $\mathcal{A}$	Yes	No
Arbitrary initial amplitude distribution	No	Yes
Number of computing nodes	2^j	$2 \leq t \leq n$
Maximum qubits at one node	$n - j$	$\max(n_0, \dots, n_{t-1})$

TABLE V. Key contrasts between the 2025 distributed amplitude amplification framework and the 2026 DEQAAA extension, adapted from [17, 18].

concrete synthesis routes such as Top-Down amplitude encoding, uniformly controlled rotations, and QGAN-based preparation circuits. Thus DEQAAA should be read not as eliminating all state-preparation bottlenecks, but as relocating them: it removes the rigid tensor-product restriction of DQAA, while replacing it with a more flexible but still nontrivial requirement of probability-aware local substate synthesis.

The empirical story in the paper is likewise slightly more nuanced than the headline percentages alone suggest. In the smallest undecomposed examples, DEQAAA does not automatically outperform centralized QAAA and EQAAA in raw gate count or depth. The practical advantage emerges after multi-controlled-gate decomposition, and it becomes markedly stronger as the qubit count grows. This is consistent with the survey-level interpretation advanced here: the principal gain is architectural and compilation-level scalability, rather than an asymptotic query-speedup over single-node exact amplification [18].

X. VARIABLE-TIME AMPLITUDE AMPLIFICATION

Standard amplitude amplification models the state-preparation routine $\mathcal{A}$ as a rigid black box. If $\mathcal{A}$ internally branches so that some computational paths halt early while others continue much longer, coherent amplification still requires padding all branches to the worst-case run-time. Denoting the maximal branch cost by T_m and the global success probability by p , the resulting complexity remains

$$\mathcal{O}\left(\frac{T_m}{\sqrt{p}}\right).$$

This worst-case synchronization overhead is the bottleneck resolved by Variable-Time Amplitude Amplification (VTAA) [11]. The obstruction is especially acute in linear-systems solvers such as HHL, where branch-dependent phase-estimation precision induces runtime spread controlled by the condition number κ .

A. Variable-Time Framework

Let a variable-time quantum procedure be specified by a sequence of stage-evolution unitaries

$$U_1, U_2, \dots, U_m,$$

and define the prefix algorithm

$$\mathcal{A}_j := U_j U_{j-1} \cdots U_1,$$

with cumulative query cost T_j satisfying $T_1 < T_2 < \cdots < T_m$. The state after stage j is $|\Psi_j\rangle = \mathcal{A}_j |0\rangle^{\otimes n}$.

To formalize early halting coherently, introduce an auxiliary status space

$$\mathcal{H}_{\text{status}} = \text{span}\{|\text{succ}\rangle, |\text{fail}\rangle, |\text{cont}\rangle\}.$$

At stage j , the joint state admits the orthogonal decomposition

$$|\Psi_j\rangle = \sqrt{p_{s,j}} |\text{succ}\rangle |\text{Good}_j\rangle + \sqrt{p_{f,j}} |\text{fail}\rangle |\text{Bad}_j\rangle + \sqrt{p_{c,j}} |\text{cont}\rangle |\Phi_j\rangle, \quad (51)$$

where $|\Phi_j\rangle$ contains branches that must continue to U_{j+1} , and $p_{s,j} + p_{f,j} + p_{c,j} = 1$.

Extending the Section I decomposition, the target space is a stage-resolved direct sum

$$\mathcal{H}_{\text{Good}} = \bigoplus_{j=1}^m \mathcal{H}_{\text{Good},j},$$

where $\mathcal{H}_{\text{Good},j}$ contains valid solutions that halt exactly at stage j . Consequently, the final prepared state can be written as

$$|\text{All}\rangle = \sum_{j=1}^m \sqrt{p_j} |\text{Good}_j\rangle + \sqrt{1-p} |\text{Bad}\rangle,$$

with

$$p = \sum_{j=1}^m p_j.$$

B. Variable-Time Amplification Limit

Theorem 4 (Variable-Time Amplification Limit [11]). *For a variable-time algorithm with stage probabilities $\{p_j\}_{j=1}^m$, cumulative costs $\{T_j\}_{j=1}^m$, and global success probability $p = \sum_{j=1}^m p_j$, there exists a variable-time amplification procedure that boosts success probability to $\Omega(1)$ using query complexity*

$$\mathcal{O}\left(T_m \log T_m + \frac{1}{\sqrt{p}} \sqrt{\sum_{j=1}^m p_j T_j^2} \cdot \log^{1.5} T_m\right).$$

The cost term $\sqrt{\sum_j p_j T_j^2}$ is the ℓ_2 -average stopping time appearing in Ambainis's analysis. When most success amplitude terminates at early stages ($T_j \ll T_m$ for dominant p_j), the second term can be substantially smaller than the worst-case synchronized cost $\mathcal{O}(T_m/\sqrt{p})$, although the additive $T_m \log T_m$ term remains.

The multiplicative $\text{polylog}(T_m)$ overhead arises from the precision requirements of coherent filtering subroutines (implemented via phase-estimation-style or polynomial filters) that separate $|\text{succ}\rangle$ from $|\text{cont}\rangle$ at each recursive level without collapsing superposition.

C. Recursive Filtering Mechanics

VTAA operates recursively by amplifying early-halting success components first. At stage $j = 1$, one applies standard amplification to increase overlap with the $|\text{succ}\rangle$ branch associated with $\mathcal{H}_{\text{Good},1}$. A coherent quantum filter then isolates branches already terminated from those in $|\text{cont}\rangle$, after which amplification is applied recursively to the continuing component at stage $j + 1$. Repeating this stage by stage prevents amplitudes that have already resolved at cost $T_1, \dots, T_j$ from being forced through the full worst-case depth T_m , which is precisely the mechanism behind the ℓ_2 runtime gain.

D. Application to Linear Systems and Bridge to QSVT

Ambainis's original motivation was the Harrow–Hassidim–Lloyd (HHL) quantum linear systems algorithm [11]. In HHL, the required simulation/phase-estimation precision is strongly controlled by the condition number κ . Replacing standard amplitude amplification with VTAA reduces the dependence from

$$\mathcal{O}(\kappa^2 \log N)$$

to

$$\mathcal{O}(\kappa \log^3 \kappa \log N),$$

providing a concrete historical proof that variable-time structure yields genuine asymptotic gains.

Conceptually, VTAA still remains inside the Grover-reflection paradigm: it optimizes temporal inhomogeneity by recursive composition of geometric amplification primitives. The next step is more radical: replacing this reflection geometry with direct polynomial transformations of singular values, leading to Section XI on Quantum Singular Value Transformation.

XI. QUANTUM SINGULAR VALUE TRANSFORMATION

A. Paradigm Shift: Geometry to Algebra

Sections IV–X analyzed amplification through one invariant plane, $\text{span}\{|\text{Bad}\rangle, |\text{Good}\rangle\}$. QSVT preserves this geometry but lifts it to a direct sum of decoupled two-dimensional planes indexed by singular values. This algebraic unification was forged through the Low–Chuang quantum signal-processing program [7, 8]. In that framework, *qubitization* is the structural step that converts a standard-form/block-encoding signal into a unitary iterate $\widehat{W}$ acting on an invariant $SU(2)$ subspace, reducing high-dimensional operator dynamics to effective single-qubit rotations [8]. The *quantum signal processor* then applies alternating, parameterized single-qubit phase shifts around those iterates; with an appropriate phase list, the induced interference synthesizes bounded polynomial transforms (including Chebyshev-type constructions) with query complexity that matches optimal polynomial-approximation limits [7, 8].

$$e^{i\phi_0 Z} \widehat{W} e^{i\phi_1 Z} \widehat{W} \dots \widehat{W} e^{i\phi_d Z}$$

QSP phase-shift sequence on a qubitized iterate $\widehat{W}$.

Gilyén et al. [6] then lifted this one-qubit mechanism to the full singular spectrum: by embedding an arbitrary (including non-Hermitian or rectangular) matrix into a unitary block encoding, QSVT applies the same QSP phase-rotation mathematics independently and in parallel to each singular value.

Let

$$A = \sum_k \sigma_k |w_k\rangle\langle v_k|$$

be the singular value decomposition of the encoded operator, and define lifted vectors

$$|\tilde{v}_k\rangle = |0\rangle^{\otimes a} |v_k\rangle, \quad |\tilde{w}_k\rangle = |0\rangle^{\otimes a} |w_k\rangle.$$

For each singular mode, the dynamics generated by U , $U^\dagger$, and ancilla-controlled phase reflections remain in a two-dimensional invariant subspace

$$\mathcal{K}_k := \text{span}\{|\tilde{v}_k\rangle, U^\dagger |\tilde{w}_k\rangle\}.$$

Hence

$$\mathcal{H}_{\text{tot}} = \bigoplus_k \mathcal{K}_k \oplus \mathcal{H}_\perp.$$

Grover's plane is therefore the rank-one special case (one relevant singular value), while QSVT executes Grover-type polynomial rotations on all singular modes simultaneously.

Amplification Framework	Suffers Overshoot?	Requires Exact p ?	Asymptotic Query Complexity
Standard Grover (Sec. IV)	Yes	Yes	$\mathcal{O}\left(\frac{1}{\sqrt{p}}\right)$
Fixed-Point AA (Sec. V)	No	No ^a	$\mathcal{O}\left(\frac{1}{\sqrt{p}} \log \frac{2}{\delta}\right)$
Oblivious AA (Sec. VI)	Yes	Yes	$\mathcal{O}\left(\frac{1}{\sqrt{p}}\right)$
Fixed-Point Oblivious (Sec. VIII)	No	No	$\mathcal{O}\left(\frac{1}{\sqrt{p}}\right)$ (average iterations, up to constants)
Distributed AA (Sec. IX)	No	No ^a	$\mathcal{O}\left(\frac{1}{\sqrt{p}} \log \frac{2}{\epsilon}\right)$ (Parallel Runtime)
Distributed Exact AA (Sec. IX)	No	Yes	exact two-phase distributed depth
Variable-Time AA (Sec. X)	No	No	$\mathcal{O}\left(T_m \log T_m + \frac{1}{\sqrt{p}} \sqrt{\sum p_j T_j^2} \log^{1.5} T_m\right)$

^a Requires only a reliable global lower bound, not the exact probabilistic value.

TABLE VI. Comparison of the core amplitude amplification frameworks developed prior to QSVT. The evolution demonstrates a systematic elimination of the "soufflé problem" (overshoot) and the relaxation of prior-knowledge constraints, culminating in optimized parallel and temporally-inhomogeneous bounds.

B. Block Encodings and the Purified Setting

Reusing the purified formalism from Section VI, let A act on s qubits and U on $a + s$ qubits. Define

$$\Pi := |0\rangle^{\otimes a} \langle 0|^{\otimes a} \otimes I_s. \quad (52)$$

Definition. U is an (α, a, ϵ) -block encoding of A if $\alpha \geq 1$ and

$$\left\| (\langle 0|^{\otimes a} \otimes I_s) U (|0\rangle^{\otimes a} \otimes I_s) - \frac{A}{\alpha} \right\| \leq \epsilon. \quad (53)$$

Equivalently, in the ancilla computational basis,

$$U = \begin{bmatrix} A/\alpha & * \\ * & * \end{bmatrix} \quad \text{up to error } \epsilon.$$

$$U \approx \left[\begin{array}{c|c} A/\alpha & * \\ \hline * & * \end{array} \right]$$

Block-encoding schematic: ancilla projection isolates the signal block A/α .

The parameter α is the subnormalization factor, a is the ancilla width, and ϵ is the synthesis error.

This notation links directly to Sections I and VI. In the canonical split $n = (n - m) + m$, the purified initialization picture isolates a signal block whose relevant singular value is $\sqrt{p}$, and

$$\mathcal{A}|0\rangle^{\otimes n} = \sqrt{p}|\text{Good}\rangle + \sqrt{1-p}|\text{Bad}\rangle, \quad (54)$$

shows that the signal singular value equals $\sqrt{p}$. Thus the initial success amplitude used throughout Grover, FPAA, oblivious amplification, and VTAA is exactly a singular-value parameter.

C. QSVT Theorem

Define the projector-controlled phase unitary

$$\Pi_\phi := e^{i\phi(2\Pi - I)}. \quad (55)$$

For a phase vector $\vec{\phi} = (\phi_0, \dots, \phi_d)$, define

$$U_{\vec{\phi}} = \Pi_{\phi_0} \prod_{j=1}^d \left(U^{(\pm)} \Pi_{\phi_j} \right), \quad (56)$$

where $U^{(\pm)}$ alternates between U and $U^\dagger$.

$\Pi_{\phi_0} U \Pi_{\phi_1} U^\dagger \Pi_{\phi_2} \dots$
 QSVT signal-processing pattern: alternating $U/U^\dagger$
 with ancilla phase controls set by $\vec{\phi}$.

Theorem 5 (QSVT Singular Value Transformation (Gilyén et al. [6])). *Let U be an (α, a, ϵ) -block encoding of*

$$A = \sum_k \sigma_k |w_k\rangle \langle v_k|,$$

and let P be a degree- d polynomial with $|P(x)| \leq 1$ on $[-1, 1]$.

If d is odd, impose odd parity $P(-x) = -P(x)$. Then there exists $\vec{\phi}$ such that

$$\left\| (\langle 0|^{\otimes a} \otimes I) U_{\vec{\phi}} (|0\rangle^{\otimes a} \otimes I) - \sum_k P(\sigma_k/\alpha) |w_k\rangle \langle v_k| \right\| \leq \mathcal{O}(d\epsilon). \quad (57)$$

If d is even, impose even parity $P(-x) = P(x)$. Then there exists $\vec{\phi}$ such that

$$\left\| (\langle 0|^{\otimes a} \otimes I) U_{\vec{\phi}} (|0\rangle^{\otimes a} \otimes I) - \sum_k P(\sigma_k/\alpha) |v_k\rangle \langle v_k| \right\| \leq \mathcal{O}(d\epsilon). \quad (58)$$

The parity condition determines output geometry: odd polynomials map right singular vectors to left singular vectors, while even polynomials act within the right-singular space.

D. Subsuming Amplitude Amplification

QSVT reproduces the phase-synthesized amplification frameworks from Sections IV–VI and VIII–X as polynomial-design instances.

1. **Standard Grover (Section IV).** For odd degree $d = 2r + 1$, the induced polynomial is

$$P_d(x) = \sin(d \arcsin x), \quad (59)$$

so $P_d(\sin \theta) = \sin(d\theta)$ (the trigonometric Chebyshev form). Amplification to constant success requires $d = \Theta(1/\sqrt{p})$.

2. **Fixed-Point Amplification (Section V).** The phase schedules derived in Section V are exact synthesis solutions for bounded minimax sign-like approximation with controlled ripple δ . The fixed-point no-overshoot property is exactly the bounded passband property of the synthesized polynomial.

Framework	Target Polynomial Behavior
Grover	$P_d(x) = \sin(d \arcsin x)$
FPAA	bounded sign-like minimax
Polynomial view of Sections IV–V: same mechanism, different approximation target.	

3. **Oblivious and Fixed-Point Oblivious Amplification (Sections VI and VIII).** In the purified model, the signal block can be written as $\sqrt{p}V$ with $V^\dagger V = I$. In block-encoding language one may either regard this as a $(1, a, 0)$ -encoding of $\sqrt{p}V$ or, equivalently, as a $(1/\sqrt{p}, a, 0)$ -encoding of V . Oblivious amplification may therefore be viewed as QSVT acting on the same singular-value parameter $\sqrt{p}$, uniformly amplifying the encoded isometry without state-dependent reflections. From this perspective, the Linear Combination of Unitaries (LCU) damping mechanism used in Fixed-Point Oblivious Amplification (Section VIII) can be interpreted as a precursor to the more general polynomial synthesis available in QSVT.
4. **Variable-Time Amplification (Section X).** VTAA’s recursive filtering is algebraically equivalent to composing polynomials that approximate a steep Heaviside selector over stage-resolved singular sectors, isolating early-halting amplitude while preserving coherence; the precision requirement yields the $\text{polylog}(T_m)$ overhead.

E. Selected Advanced QSVT Instantiations

Beyond the direct unification statements above, several high-impact algorithmic primitives become immediate once amplitude amplification is interpreted as polynomial synthesis over singular values:

1. **Spectral thresholding and filtering.** By synthesizing bounded approximants to step-like functions, QSVT implements coherent singular-value filtering, enabling selective amplification or suppression of spectral bands without intermediate measurement [6].
2. **Matrix inversion from polynomial reciprocals.** Choosing $P(x) \approx 1/x$ on a gapped interval $[1/\kappa, 1]$ recovers linear-system subroutines in the same algebraic framework, clarifying why condition-number dependence is governed by approximation degree rather than bespoke circuit geometry [6, 11].
3. **Stable amplification inside block-encoding pipelines.** The fixed-point intuition of Sections V and VI lifts to robust singular-value shaping: instead of tuning iteration counts branch-by-branch, one compiles a single bounded polynomial with explicit passband and stopband behavior.

XII. APPLICATIONS OF THE UNIFIED FRAMEWORK

By shifting the focus from geometric state rotations to singular-value polynomial transforms, the amplification framework extends naturally to other quantum computational primitives. Because the structure applies bounded polynomials to matrix singular blocks, it connects amplitude amplification to broader classes of algorithms beyond unstructured search.

Linear Systems. Choosing $P(x) \approx 1/x$ over an interval $[1/\kappa, 1]$ yields quantum linear-system solvers. This recovers the condition-number scaling improvements discussed in the context of the HHL algorithm [2, 11]. Both amplifying marked amplitude and inverting spectral amplitude reduce to synthesizing a bounded polynomial with the requisite approximation profile.

Hamiltonian Simulation. Simulating quantum time evolution, e^{-iHt} , reduces to applying trigonometric polynomials $P(x) = \cos(tx)$ and $P(x) = \sin(tx)$ to a block-encoded Hamiltonian H . If H is given via an (α, a, ϵ) -block encoding, this framework yields the near-optimal scaling $\mathcal{O}(\alpha t + \log(1/\epsilon))$ without relying on Trotterization or truncated Taylor series expansions.

Ground State Preparation (Beyond VQE and QAOA). In the current Noisy Intermediate-Scale Quantum (NISQ) era, ground state preparation and optimization are dominated by variational heuristics such as the Variational Quantum Eigensolver (VQE) and the Quantum Approximate Optimization Algorithm (QAOA). While useful for near-term hardware, these heuristic algorithms lack rigorous asymptotic performance guarantees. In the fault-tolerant setting, QSVT offers a more systematic alternative to these methods. By applying a polynomial approximation of the sign function or the exponential function to a block-encoded Hamiltonian, one

can project an initial trial state toward the ground state with provable, near-optimal query complexity.

Amplitude Estimation. Quantum Amplitude Estimation (QAE), introduced by Gilles Brassard, Peter Høyer, Michele Mosca, and Alain Tapp in the same foundational work that formalized generalized amplitude amplification [9], is the natural twin primitive for estimating success probabilities rather than only amplifying them. The original construction combines Grover-style iterations with Shor-style Fourier analysis: for the generalized iterate Q with eigenvalues $e^{\pm i2\theta_a}$ (where $a = \sin^2 \theta_a$ is the target success probability), QAE applies controlled powers of Q and the size- M Fourier transform F_M to estimate θ_a , then recovers a . This coherent “probability counting” capability made QAE a central subroutine in quantum Monte Carlo methods for finance and in quantum machine-learning workflows. The main implementation bottleneck is the explicit Quantum Fourier Transform (QFT), which is costly in ancilla and depth. Modern QSP/QSVT-based formulations bypass that bottleneck by using bounded singular-value polynomials that directly extract phase/amplitude information without an explicit QFT, substantially reducing overhead and making polynomial-based amplitude estimation more favorable for NISQ and early fault-tolerant architectures [6, 7, 10].

Quantum Machine Learning. Matrix inversion and projection are central bottlenecks in classical machine learning. Applying appropriate thresholding or inverse polynomials to a data matrix block-encoding provides a native mechanism for quantum principal component analysis (PCA) and regression.

XIII. LIMITS AND OPTIMALITY

Theorem 6 (BBBV Lower Bound [3]). *Let $f : \{0,1\}^n \rightarrow \{0,1\}$ be a black-box Boolean oracle with $N = 2^n$ inputs and $M = |f^{-1}(1)| \geq 1$ marked items. Any quantum query algorithm that finds a marked input with success probability at least $2/3$ (equivalently, any fixed constant bounded away from 0) must use*

$$\Omega\left(\sqrt{N/M}\right)$$

oracle queries in the worst case.

This theorem provides the standard optimality benchmark for amplitude amplification in the unstructured oracle model: Grover-type scaling is not merely sufficient, but also necessary up to constant factors.

Proof. We present a self-contained hybrid proof in the phase-oracle model following the Bennett–Bernstein–Brassard–Vazirani argument [3]. Let $S \subseteq [N]$ denote the marked set, with $|S| = M$, and define the phase oracle

$$O_S |i\rangle |z\rangle = (-1)^{[i \in S]} |i\rangle |z\rangle,$$

where the second register $|z\rangle$ contains arbitrary workspace. A T -query algorithm has the form

$$U_T O_S U_{T-1} O_S \cdots U_1 O_S U_0 |0\rangle,$$

where the U_t are oracle-independent unitaries.

We first reduce search to decision. Suppose an algorithm $\mathcal{A}$ finds a marked item with probability at least $2/3$ using T queries whenever $|S| = M$. Construct a decision algorithm $\mathcal{D}$ with $T+1$ queries as follows: run $\mathcal{A}$, measure its output candidate i , query the oracle once on i , and accept iff $i \in S$. Then:

1. if $|S| = M$, the acceptance probability of $\mathcal{D}$ is at least $2/3$;
2. if $S = \emptyset$, the acceptance probability of $\mathcal{D}$ is 0.

Therefore it suffices to prove that any algorithm distinguishing the empty oracle from every oracle with exactly M marked items and constant bias must use $\Omega(\sqrt{N/M})$ queries.

Let $\mathcal{D}$ be such a decision algorithm using T queries. For each marked set S , define the final state

$$|\psi_T^S\rangle = U_T O_S U_{T-1} O_S \cdots U_1 O_S U_0 |0\rangle,$$

and let

$$|\psi_T^\emptyset\rangle = U_T U_{T-1} \cdots U_1 U_0 |0\rangle$$

be the final state under the empty oracle $O_\emptyset = I$.

To compare these states, introduce the intermediate empty-oracle states

$$|\phi_t\rangle = U_t U_{t-1} \cdots U_0 |0\rangle, \quad t = 0, 1, \dots, T-1.$$

Write the query register decomposition of $|\phi_t\rangle$ as

$$|\phi_t\rangle = \sum_{i=1}^N |i\rangle |\eta_{t,i}\rangle,$$

where the workspace vectors $|\eta_{t,i}\rangle$ need not be normalized. Define

$$q_t(i) = \|\eta_{t,i}\|^2, \quad \sum_{i=1}^N q_t(i) = 1.$$

Thus $q_t(i)$ is exactly the probability that the t -th query would address index i if all oracle calls were replaced by the identity.

The effect of changing the empty oracle into O_S at this step is

$$(O_S - I) |\phi_t\rangle = -2 \sum_{i \in S} |i\rangle |\eta_{t,i}\rangle,$$

so

$$\|(O_S - I) |\phi_t\rangle\| = 2 \sqrt{\sum_{i \in S} q_t(i)}.$$

To make the hybrid step explicit, define for $t = 0, 1, \dots, T$ the final hybrid states

$$|\chi_t\rangle = U_T O_S U_{T-1} O_S \cdots U_{t+1} O_S U_t U_{t-1} \cdots U_0 |0\rangle,$$

so that $|\chi_0\rangle = |\psi_T^S\rangle$ and $|\chi_T\rangle = |\psi_T^\emptyset\rangle$. Then, for each $t = 0, 1, \dots, T-1$,

$$|\chi_t\rangle - |\chi_{t+1}\rangle = U_T O_S \cdots U_{t+2} O_S U_{t+1} (O_S - I) |\phi_t\rangle.$$

Because left multiplication by a unitary preserves norm,

$$\| |\chi_t\rangle - |\chi_{t+1}\rangle \| = \| (O_S - I) |\phi_t\rangle \|.$$

Applying the triangle inequality to the telescoping sum

$$|\psi_T^S\rangle - |\psi_T^\emptyset\rangle = \sum_{t=0}^{T-1} (|\chi_t\rangle - |\chi_{t+1}\rangle)$$

therefore yields

$$\| |\psi_T^S\rangle - |\psi_T^\emptyset\rangle \| \leq \sum_{t=0}^{T-1} \| (O_S - I) |\phi_t\rangle \| = 2 \sum_{t=0}^{T-1} \sqrt{\sum_{i \in S} q_t(i)}.$$

Average this bound over all M -element subsets $S \subseteq [N]$. For each fixed t ,

$$\mathbb{E}_S \left[\sum_{i \in S} q_t(i) \right] = \sum_{i=1}^N q_t(i) \Pr[i \in S] = \frac{M}{N} \sum_{i=1}^N q_t(i) = \frac{M}{N}.$$

Because $x \mapsto \sqrt{x}$ is concave, Jensen's inequality gives

$$\mathbb{E}_S \left[\sqrt{\sum_{i \in S} q_t(i)} \right] \leq \sqrt{\mathbb{E}_S \left[\sum_{i \in S} q_t(i) \right]} = \sqrt{\frac{M}{N}}.$$

Hence

$$\mathbb{E}_S [\| |\psi_T^S\rangle - |\psi_T^\emptyset\rangle \|] \leq 2T \sqrt{\frac{M}{N}}.$$

Therefore there exists at least one marked set S with $|S| = M$ such that

$$\| |\psi_T^S\rangle - |\psi_T^\emptyset\rangle \| \leq 2T \sqrt{\frac{M}{N}}.$$

Now let Π be the POVM element corresponding to the algorithm declaring “marked.” By assumption,

$$\langle \psi_T^S | \Pi | \psi_T^S \rangle \geq \frac{2}{3}, \quad \langle \psi_T^\emptyset | \Pi | \psi_T^\emptyset \rangle \leq \frac{1}{3}.$$

Define

$$\rho_S = |\psi_T^S\rangle \langle \psi_T^S|, \quad \rho_\emptyset = |\psi_T^\emptyset\rangle \langle \psi_T^\emptyset|.$$

For pure states, the bias achievable by any measurement is upper-bounded by their trace distance, and for pure states that trace distance is at most the Euclidean distance:

$$\begin{aligned} |\langle \psi_T^S | \Pi | \psi_T^S \rangle - \langle \psi_T^\emptyset | \Pi | \psi_T^\emptyset \rangle| &\leq \frac{1}{2} \|\rho_S - \rho_\emptyset\|_1 \\ &\leq \| |\psi_T^S\rangle - |\psi_T^\emptyset\rangle \|. \end{aligned} \quad (60)$$

Since the left-hand side is at least $1/3$, we conclude that

$$\frac{1}{3} \leq \| |\psi_T^S\rangle - |\psi_T^\emptyset\rangle \| \leq 2T \sqrt{\frac{M}{N}}.$$

Thus

$$T \geq \frac{1}{6} \sqrt{\frac{N}{M}}.$$

Recalling the initial reduction from search to decision, any search algorithm must satisfy

$$T + 1 = \Omega\left(\sqrt{\frac{N}{M}}\right),$$

and therefore $T = \Omega(\sqrt{N/M})$ as claimed. $\square$

A. Alternative Lower-Bound Lenses

Polynomial method (Beals et al.). If a quantum algorithm makes T oracle queries, then each final-state amplitude is a complex multilinear polynomial of degree at most T in the oracle input bits, so the acceptance probability has degree at most $2T$ [4]. For search, taking the success amplitude from order $\sqrt{M/N}$ to a constant forces degree $\Omega(\sqrt{N/M})$ by classical approximation theory.

Adversary method (Ambainis). The adversary framework packages the same obstruction into a progress argument on distinguishability between different oracle instances [5]. For unstructured search it reproduces the same $\Omega(\sqrt{N/M})$ lower bound, but in a form that often generalizes more flexibly to other oracle problems.

B. From Lower-Bound Tool to Algorithmic Blueprint

For many years, the polynomial method was primarily interpreted as a barrier argument describing what quantum algorithms cannot do. QSVT changes this perspective by providing a constructive compilation theorem showing that bounded parity-constrained polynomials can be implemented as explicit quantum circuits on block-encoded operators (Section XI). In this sense, modern amplification not only respects polynomial lower bounds, but also approaches the boundary they permit from a constructive direction.

C. Scope of the Bound: Unstructured Versus Structured Problems

The BBBV statement is specific to *unstructured* black-box search. If additional algebraic or combinatorial structure is available, the effective query model changes and

stronger speedups can emerge. This is the canonical contrast between Grover and Shor: Grover gives quadratic improvement for blind search/inversion, whereas Shor achieves polynomial-time factoring by exploiting periodic and group-theoretic structure. Thus, the $\Omega(\sqrt{N})$ barrier is universal for unstructured search, but it is not a universal barrier for all quantum computation.

XIV. CONCLUSION AND OUTLOOK

The progression from geometric reflections to singular value transformations provides a mathematically cohesive framework for amplitude amplification. Algorithm design in this regime shifts from heuristic circuit construction to the identification of bounded polynomials.

While the asymptotic theory within the oracle model is mature, the physical realization of these frameworks presents several active research frontiers:

- **Fault-Tolerant Compilation:** The phase-modulated reflection sequences require deep multi-controlled operations that are costly to compile into

fault-tolerant discrete gate sets (e.g., Clifford+T).

- **Distributed Architectures:** As outlined in Section IX, executing massive block-encodings across modular, distributed NISQ networks remains a critical, unsolved hardware challenge.
- **Algorithmic Automation:** Because the framework reduces algorithm design to polynomial synthesis, optimal phase angles can be discovered programmatically via classical optimization algorithms (such as the Remez algorithm), enabling the automated generation of quantum circuits for specific approximation targets.
- **Efficient Data Loading:** Constructing the requisite block-encodings with realistic quantum RAM (QRAM) and memory assumptions remains a practical bottleneck.

Amplitude amplification has evolved from an isolated geometric trick into a standardized application of polynomial operator calculus, shifting the focus of future research from query complexity to hardware implementation.

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